

Morphological Differences of Punjab (Mastuana) and Himachal Pradesh (Bilaspur) STC Boxers Ranging from Age 16 to 20 years.

Jaswinder Singh¹, Y.Momo Singh¹ & S.S.Kang¹

ABSTRACT

The present anthropometric study was conducted during September-October 2017 on 31 male STC boxers of Punjab and Himachal Pradesh. Anthropometric measurements were taken following standard techniques of Ross et al. (1978) with standard instruments. Somatotype was computed through computer equations of Carter (1980). Percent body fat was assessed from body density by following equations of Durnin & Womersley (1974) and Brozek et al. (1963). Mean, standard deviation and student's t-test was used for comparison of two groups. The results of study indicate height, weight, sitting height, endomorphic and mesomorphic component of somatotype shows non-significant differences between two groups of boxers. Height weight ratio and ectomorphy values were significantly higher in 17 year group of Himachal Pradesh boxers than Punjab. The findings of study were compared with Indian Junior and senior boxers of 2006.

INTRODUCTION

Morphological differences are important to find out champion athletes. It's well known that athletes are leaner and muscular than non-athletes. Lot of bodily changes occurs due to exercise in sportsmen compared to sedentary person. These changes in body reflect through external and internal features of person. In this regard many studies have been conducted on athletes all over the World. Sodhi & Sidhu in 1984 published a book "Physique & Selection of Sportsmen." Tanner (1964) published a book "Physique of Olympic Athletes after conducting study on Olympians. De Garry et al. (1974) published research work on

Anthropological and Genetics studies on Olympians. Somatotype, body composition and body proportionality studies on Olympians were also published by Carter et al. (1984). Somatotype and sport selection research work have been reported by Carter 1990. Thus lots of studies indicate most of athletes resembles in morphological parameters like somatotype and body composition of same games. More than 50% Olympic athletes of different games resembles in physique as reported by Carter (1990). Therefore, morphological studies are very important to know body differences among athletes of same game as well as different games. Keeping this in mind this study has been conducted to observe body

¹ Junior Scientific Officer, Department Sports Anthropometry, NSNIS Patiala.

morphology of Punjab boxers and Himachal boxers of STC Scheme of Sports Authority of India. This study will provide age-wise, weight category-wise difference in morphological parameters. The study will provide baseline research work regarding anthropometric parameters in boxing.

MATERIAL AND METHODS:

The present cross-sectional study has been conducted on 14 male Punjab and 17 male Himachal Pradesh STC boxers. The area of data collection was Mastuana Sahib in Punjab and Bilaspur in Himachal Pradesh (Sports Authority Training Centers) during months of September and October, 2017. Anthropometric measurements were taken by following standard techniques of Ross et al. (1978) using standard instruments. Somatotype components (endomorph, mesomorph, ectomorph and height weight ratio) were calculated according to computer equations of Carter (1980). Body density was assessed by equations of Durnin and Womersley (1974). Percent body fat was calculated using body density by formulae of Brozek et al. (1963). Mean and standard deviations were evaluated through Microsoft Excel computer programme. Student's t-test was used for comparison between two groups.

RESULTS AND DISCUSSION

Table 1 shows the height, body weight and sitting height differences in Punjab (Mastuana) and Himachal Pradesh (Bilaspur) STC boxers ranging in age from 16 to 20 years. Both groups did not differ in age as indicated by test of significance. The boxers of Punjab were found to be more than 5 cm shorter at 17 years than Himachal Pradesh. But the results of height between two groups were statistically non-significant. Variations in standard deviations between above two groups were found to be more at 17 year. So, these groups represent

almost average stature of their population. The age-wise trend in body weight was not uniform due to weight category sport. In Punjab sample minimum body weight was 60kg in 18-20 year group whereas 62.96kg in 17th year age group of Himachal Pradesh. The findings of body weight were non-significant in both groups age-wise due to large variations in term of Standard deviation. Himachal Pradesh boxers were found to be lighter for height than Punjab boxers but significant difference occurs at 17 years of age ($t=2.98$). Sitting height differences were found to be statistically non-significant between these two groups. However, at 18-20 years group Himachal boxers were about 3.18 cm shorter than Punjab boxers. But difference in height was 0.48cm. Thus, lower proportion of Punjab boxers was shorter than Himachal Pradesh boxers in this age group.

Mean values of bony diameters of humerus, wrist, femur and ankle were shown in Table 2. Punjab and Himachal Pradesh boxers differs slightly each other. In statistical terms both groups differs non-significantly except ankle diameters at 17 year of age, former have lesser value than latter. Thus, in general both groups resembles in bony diameters. Table 3 shows the Circumference differences in Punjab (Mastuana) and Himachal Pradesh (Bilaspur) STC boxers. Generally Himachal Pradesh boxers have slightly more circumference values of upper arm, forearm, thigh and calf. But the statistically differences have not been found due to non-homogeneous sample. Such results may be due to small data size, age-wise distribution of sample, In each age group different weight categories clubbed together. Moreover both groups represents near to adult values of population. Himachal Pradesh boxers have significantly more

developed thighs than Punjab boxers in 18-20 year group. The body weight of former has been found about 5.74kg lesser than latter. Thus, development of circumferences in Himachal Pradesh boxers shows similar trends to body weight.

The results of skinfolds were shown in Table 4. In case of biceps, triceps and forearm skinfolds Himachal Pradesh boxers were leaner in skinfold thickness but significant difference occurs only in biceps at 17 year group. The subscapular, suprailiac, supraspinale & abdominal skinfold values were more in Himachal Pradesh boxers at 18-20 year group than Punjab boxers. But difference in these skinfolds were non-significant (subscapular, suprailiac, supraspinale & abdominal skinfold). The thigh and calf skinfolds values were significantly higher at 16 year in Punjab boxers than Himachal Pradesh boxers. But the differences were non-significant in thigh and calf skinfold at 17 and 18-20 year age between these two groups. The boxers of Punjab were fatter than Himachal Pradesh at 16 and 17 years of age. But statistically differences were non-significant due to higher standard deviation in Punjab sample. This may be due to large variations found (SD) in body fat due to age-wise study and small sample size and weight categories grouped together.

Table 5 shows the Somatotypic differences in Punjab (Mastuana) and Himachal Pradesh (Bilaspur) STC boxers. STC boxers of Punjab were more endomorphic at 16 and 17 years of age than Himachal Pradesh boxers. However, both groups did not differ in endomorphic component of somatotype. Similar trend has been found in development of mesomorphy. The ectomorphic component has been found lesser in Punjab boxers than

latter at 16 and 17 years of age. Difference was statistically significant at 17 year. Thus, large variations occur in somatotype due to different weight categories in a same age group. This may be the reason non-significant differences in ectomorphy at 16, 18 to 20 year group. The variation in standard deviation was lesser in 16 year group and differences were significant in ectomorphy. So, there were large differences in body dimensions and somatotypes in boxers age wise.

Table 6 shows the anthropometric parameters of 48 to 50kg weight category. The sample size was 7 for Punjab boxers whereas only two in case of Himachal Pradesh boxers. The Himachal Pradesh boxers were significantly taller and more ectomorphic than STC male Punjab boxers. Both groups resemble in sitting height. The lower extremities of Himachal boxers were taller than Punjab boxers. The boxers of Himachal Pradesh have lesser fat than Punjab in 48 to 50kg weight category but findings were not statistically supported. Such type of findings may be due to smaller sample size and large variation among players as indicated by standard deviation.

As shown in Table 7, 52 to 56 kg weight class STC boxers of Punjab were significantly fatter and more endomorphic than Himachal Pradesh STC boxers. However, both groups did not differ in height, weight, sitting height, mesomorphic and endomorphic components of somatotype.

Table 8 shows the anthropometric parameters of 57-60kg weight category boxers of Punjab and Himachal Pradesh. Punjab STC boxers were about 1 year younger, 2cm taller and 0.5kg lighter in weight category 57-60kg than Himachal Pradesh STC boxers. Punjab boxers have

taller trunk length, more endomorphic, lesser mesomorphic, higher ectomorphic rating than Himachal boxers. Punjab STC boxers were 5 % more fatty than Himachal Pradesh boxers. This may be due to differences in altitude as well as genetics reasons between two groups. In statistical terms, sitting height, endomorphy and body fat were found significantly more in Punjab STC boxers than Himachal STC boxers of Bilaspur.

Table 9 depicts morphological differences between Punjab and Himachal Pradesh STC boxers of 63 to 66 kg weight category. The average age of Punjab boxers at the time of examination was about two year more than Himachal Pradesh boxers. Punjab boxers were shorter in height and sitting height than Himachal Pradesh STC boxers. Punjab boxers were proportionally heavier, more muscular and one percent lesser fatty than Himachal Pradesh boxers. However, significant differences have been noticed in height, height-weight ratio, mesomorphy and ectomorphy between these two groups. Height proportional weight is lesser in Himachal Pradesh boxers compared to Punjab boxers. But sample size in limited to two only in Himachal and 3 in Punjab boxing STC. More sample size required to establish differences in body morphology.

Table 10 shows the morphological features of 67 to 70 kg weight class of Punjab STC and Himachal STC boxers. Punjab boxers of this category were younger in age, taller having larger trunk and lesser mesomorphic than Himachal boxers. However, significant differences occurs only in mesomorphy, the former being lesser mesomorphic than latter. The study concludes in this weight category, both groups differ only in mesomorphy. The

differences in mean age, height and sitting height may be non-significant due to higher standard deviations and lesser sample size in this weight category.

Table 11 shows that Himachal STC boxers were significantly older than Punjab boxers in weight category 80kg group. However, boxers of Punjab were taller having longer trunks, lower musculo-skeletal development per unit height, fatter than Himachal Pradesh boxers but results not statistically supported. Both groups were heavier for height also percent fat was high in boxers. Thus, on basis of mean values findings are not realistic as wide variation occurs among players due to small sample size. Boxers were advised to reduce fat.

As shown in Table 12 the present study Punjab boxers at age 16 & 17 year were about 2-3cm shorter than Indian Junior boxers whereas at 18-20 yr about 10cm shorter than senior Indian boxers reported by Khanna & manna (2006). It is interesting to observe 16 & 17 year Punjabi boxers of present study were about 10kg heavier than Indian junior boxers. However they (18-20 yrs Punjab Boxers) were about 10 cm shorter and about 16kg lighter than Senior Indian boxers. The Boxers of Punjab at 16 & 17 years of age were more endomorphic, mesomorphic and lesser ectomorphic than Junior boxers of Khanna & Manna (2006). The 18-20 year Punjab boxer shows lesser endomorphy and higher ectomorphy compared to Senior Indian boxers. The percent fat has been found higher in 16, 17 years Punjabi boxers than Indian junior boxers. However, 18-20 year Punjabi boxers were found to be leaner (Fat 10.23%) than Indian senior boxers (fat 16.4%).

The STC boxers of Himachal Pradesh boxers were more mesomorphic than Junior and Senior boxers studied by Khanna and

Manna (2006). Himachal boxers of 18-10 year of age were shorter in height than Indian seniors. But body weight of 16 and 17 year Himachal boxers were more than Junior boxers at similar height whereas former were lighter and shorter at 18-20 year compared to Indian seniors (Table 12). This study highlights age-wise and weight category-wise differences in body morphology. The development of mesomorphy is better than Indian juniors and equivalents to Indian seniors. The body

weight proportional to height was lesser than Indian senior boxers but more than Indian juniors. This may be due to differential maturity status among juniors or age manipulations. Moreover large data required to know morphological differences in boxers. There were wide differences in height body weight and other anthropometric parameters as indicated by standard deviations. This study will provide baseline anthropometric information regarding STC boxers of Punjab and Himachal Pradesh.

Table-1 : Height, body weight and sitting height differences in Punjab (Mastuana) and Himachal Pradesh (Bilaspur) STC boxers.

Parameter	Age (yrs)	Punjab (Mastuana Sahib)			Himachal Pradesh (Bilaspur)			Statistics
		N	Mean	SD	N	Mean	SD	t-value
Age Group	16	5	16.08	0.31	4	15.78	0.21	1.66
	17	4	16.78	0.38	8	16.92	0.32	0.67
	18-20	5	18.86	0.99	5	19.26	1.0	0.62
Height (cm)	16	5	172.06	7.75	4	172.25	8.35	0.04
	17	4	169.47	11.19	8	174.85	5.47	0.91
	18-20	5	169.38	5.03	5	168.9	2.14	0.20
Weight (kg)	16	5	66.38	13.74	4	59.47	5.68	1.02
	17	4	66.78	19.07	8	62.96	8.01	0.38
	18-20	5	60.0	1.85	5	65.74	11.17	1.13
HWR	16	5	42.74	1.72	4	44.15	1.33	1.38
	17	4	42.10	0.98	8	44.05	1.22	2.98*
	18-20	5	43.32	0.85	5	42.05	2.26	1.18
Sitting height (cm)	16	5	89.36	2.46	4	89.95	4.75	0.22
	17	4	88.62	4.79	8	90.75	2.32	0.84
	18-20	5	87.66	2.70	5	84.48	4.55	0.08

Table-2

Bony diameter differences in Punjab (Mastuana) and Himachal Pradesh (Bilaspur) STC boxers.

Parameter	Age (yrs)	Punjab (Mastuana Sahib)			Himachal Pradesh (Bilaspur)			Statistics
		N	Mean	SD	N	Mean	SD	t-value
Biepicondylarhumerus(cm)	16	5	6.84	0.56	4	6.77	0.42	0.20
	17	4	6.77	0.52	8	7.0	0.37	0.76
	18-20	5	6.84	0.20	5	6.94	0.45	0.45
Wrist (cm)	16	5	5.40	0.43	4	5.65	0.24	1.10
	17	4	5.55	0.70	8	5.75	0.31	0.54
	18-20	5	5.58	0.15	5	5.84	0.52	1.07
Biepicondylar femur (cm)	16	5	9.76	0.72	4	9.7	0.36	0.16
	17	4	9.90	0.69	8	9.95	0.45	0.13
	18-20	5	9.60	0.11	5	9.86	0.78	0.73
Ankle (cm)	16	5	7.04	0.35	4	7.1	0.42	0.22
	17	4	6.75	0.66	8	7.45	0.24	2.03*
	18-20	5	6.90	0.17	5	7.14	0.44	1.12

Table : 3

Circumference differences in Punjab (Mastuana) and Himachal Pradesh (Bilaspur) STC boxers.

Parameter	Age (yrs)	Punjab (Mastuana Sahib)			Himachal Pradesh (Bilaspur)			Statistics
		N	Mean	SD	N	Mean	SD	t-value
Upper arm relaxed (cm)	16	5	27.64	4.09	4	27.65	2.73	0
	17	4	27.77	3.11	8	27.25	2.19	0.30
	18-20	5	27.16	0.85	5	29.38	3.0	1.59
Upper arm flexed (cm)	16	5	30.84	3.92	4	30.85	2.60	0
	17	4	31.10	3.39	8	30.30	1.89	0.43
	18-20	5	31.26	3.39	5	31.82	2.80	0.40
Forearm (cm)	16	5	26.06	2.91	4	25.77	0.98	0.20
	17	4	26.30	2.80	8	26.14	1.36	0.11
	18-20	5	26.10	1.13	5	27.0	2.62	0.70
Thigh (cm)	16	5	48.06	4.25	4	48.77	1.95	0.33
	17	4	51.85	7.58	8	50.43	4.02	0.35
	18-20	5	47.22	2.53	5	53.86	6.27	2.19*
Calf (cm)	16	5	34.38	3.58	4	33.30	1.75	0.59
	17	4	34.52	3.62	8	34.47	2.40	0.02
	18-20	5	33.14	1.65	5	35.56	3.82	1.29
Wrist(cm)	16	5	16.42	1.30	4	16.80	0.95	0.33
	17	4	16.18	1.98	8	16.59	0.42	0.41
	18-20	5	16.22	0.47	5	16.86	1.14	1.16
Ankle (cm)	16	5	21.26	2.22	4	21.10	1.0	0.14
	17	4	21.18	1.84	8	21.52	1.19	0.34
	18-20	5	21.0	1.06	5	21.20	1.58	0.23

Table - 4 :
Skinfold and percent fat differences in Punjab (Mastuana) and Himachal Pradesh
(Bilaspur) STC boxers.

Parameter	Age (yrs)	Punjab (Mastuana Sahib)			Himachal Pradesh (Bilaspur)			Statistics
		N	Mean	SD	N	Mean	SD	
Biceps (mm)	16	5	3.52	0.64	4	3.1	0.11	1.43
	17	4	5.90	2.18	8	3.25	0.48	2.39*
	18-20	5	3.92	1.20	5	3.68	1.75	0.25
Triceps (mm)	16	5	10.96	6.82	4	8.15	1.63	0.89
	17	4	12.50	7.81	8	8.08	2.53	1.10
	18-20	5	6.04	1.70	5	7.88	5.23	0.74
Forearm(mm)	16	5	7.32	3.25	4	4.5	0.57	1.90
	17	4	7.30	3.53	8	5.3	0.91	1.11
	18-20	5	5.24	1.70	5	4.72	2.28	0.41
Subscapular(m m)	16	5	12.28	6.26	4	8	1.47	1.47
	17	4	14.50	11.27	8	6.82	1.67	1.35
	18-20	5	7.72	3.25	5	9.80	4.39	0.85
Suprailiac (mm)	16	5	13.24	8.93	4	6.95	1.77	1.53
	17	4	15.40	14.10	8	6.20	1.14	1.30
	18-20	5	5.76	3.81	5	9.48	7.05	1.03
Supraspinale(m m)	16	5	7.68	4.56	4	4.35	0.1	1.63
	17	4	8.80	6.44	8	4.30	0.89	1.38
	18-20	5	4.68	0.80	5	5.40	3.05	0.54
Abdominal(mm)	16	5	15.88	10.88	4	7.65	1.29	1.67
	17	4	17.87	15.18	8	8.30	2.03	1.25
	18-20	5	8.88	3.97	5	12.12	7.32	0.83
Thigh(mm)	16	5	15.96	8.0	4	8.05	1.49	2.16*
	17	4	16.95	10.40	8	9.32	2.82	1.44
	18-20	5	6.12	2.30	5	11.32	6.53	1.67
Calf(mm)	16	5	11.18	5.45	4	5.2	0.99	2.36*
	17	4	10.90	5.77	8	6.07	2.17	1.61
	18-20	5	5.04	2.25	5	6.0	6.53	0.55
Body fat (%)	16	5	15.28	5.94	4	11.77	1.51	1.27
	17	4	16.89	6.97	8	10.88	2.10	1.68
	18-20	5	10.23	4.95	5	12.46	5.31	0.68

Table -5

Somatotypic differences in Punjab (Mastuana) and Himachal Pradesh (Bilaspur) STC boxers.

Parameter	Age (yrs)	Punjab (Mastuana Sahib)			Himachal Pradesh (Bilaspur)			Statistics
		N	Mean	SD	N	Mean	SD	t-value
Endomorphy	16	5	2.98	1.70	4	1.94	0.20	1.35
	17	4	3.40	2.05	8	1.76	0.34	1.58
	18-20	5	1.74	0.54	5	2.23	1.35	0.76
Mesomorphy	16	5	4.64	1.46	4	4.50	1.14	0.16
	17	4	5.05	0.50	8	4.58	1.08	1.05
	18-20	5	4.97	0.82	5	5.71	1.72	0.90
Ectomorphy	16	5	2.70	1.26	4	3.74	0.97	1.39
	17	4	2.23	0.71	8	3.66	0.89	2.99*
	18-20	5	3.14	0.62	5	2.30	1.47	1.16

Table 6

Weight category wise morphological profile of Punjab and Himachal Pradesh STC boxers (Weight category 48-50 kg, N= 9)

Parameter	Mastuana (n = 7)		Bilaspur (n = 2)		Statistics
	Mean	SD	Mean	SD	t-value
Decimal Age (yrs.)	15.93	1.48	15.20	0.63	1.02
Height (cm)	163.83	4.15	168.10	0.14	2.72*
Weight (kg)	52.19	2.43	51.00	2.26	0.65
Sitting Height	86.00	1.63	85.90	0.42	0.15
Height Weight Ratio	43.85	1.15	45.34	0.71	2.24*
Endomorphy	2.06	0.69	1.23	0.89	1.22
Mesomorphy	4.04	0.86	3.58	0.04	1.41
Ectomorphy	3.52	0.84	4.61	0.52	2.24*
Fat (%)	11.40	2.48	8.15	5.35	0.83

Table 7

Weight category wise morphological profile of Punjab and Himachal Pradesh STC boxers (Weight category 52-56 kg, N= 8)

Parameter	Mastuana (n=4)		Bilaspur (n=4)		Statistics
	Mean	SD	Mean	SD	t-value
Decimal Age (yrs.)	16.96	1.81	18.03	1.52	0.91
Height (cm)	168.95	3.05	169.90	4.89	0.33
Weight (kg)	57.98	4.17	55.63	3.34	0.88
Sitting Height	87.55	2.36	88.45	3.29	0.44
Height Weight Ratio	43.70	1.49	44.52	0.88	0.95
Endomorphy	1.96	0.23	1.47	0.31	2.54*
Mesomorphy	4.34	1.71	4.18	0.48	0.18
Ectomorphy	3.40	1.09	4.01	0.64	0.96
Fat (%)	12.07	1.44	9.04	1.59	2.82*

Table 8:

Weight category wise morphological profile of Punjab and Himachal Pradesh STC boxers (Weight category 57-60 kg, N= 8)

Parameter	Mastuana (n=2)		Bilaspur (n=6)		Statistics
	Mean	SD	Mean	SD	t-value
Decimal Age (yrs.)	15.92	0.87	17.15	1.72	1.32
Height (cm)	172.80	5.23	170.80	5.09	0.47
Weight (kg)	60.50	3.11	61.02	1.56	0.23
Sitting Height	91.15	0.21	89.50	1.87	2.12*
Height Weight Ratio	44.04	2.09	43.39	1.33	0.41
Endomorphy	2.30	0.05	1.74	0.23	5.58*
Mesomorphy	4.33	1.44	5.18	0.83	0.79
Ectomorphy	3.66	1.53	3.18	0.97	0.42
Fat (%)	15.17	0.31	10.60	1.05	9.49*

Table 9

Weight category wise morphological profile of Punjab and Himachal Pradesh STC boxers (Weight category 63-66 kg, N= 5)

Parameter	Mastuana (n=3)		Bilaspur (n=2)		Statistics
	Mean	SD	Mean	SD	t-value
Decimal Age (yrs.)	18.26	2.70	16.35	0.36	1.21
Height (cm)	173.73	1.06	181.85	2.90	3.79*
Weight (kg)	65.47	2.76	65.10	1.98	0.17
Sitting Height	89.00	0.53	93.30	4.95	1.22
Height Weight Ratio	43.12	0.38	45.21	0.26	7.30*
Endomorphy	2.07	0.96	2.02	0.29	0.08
Mesomorphy	5.03	0.24	3.27	0.60	3.94*
Ectomorphy	2.98	0.28	4.51	0.19	7.28*
Fat (%)	11.30	6.12	12.32	2.13	0.27

Table 10

Weight category wise morphological profile of Punjab and Himachal Pradesh STC boxers (Weight category 67-70 kg, N= 4)

Parameter	Mastuana (n=2)		Bilaspur (n=2)		Statistics
	Mean	SD	Mean	SD	t-value
Decimal Age (yrs.)	15.70	1.09	18.42	1.74	-1.87
Height (cm)	176.65	3.04	172.85	1.91	1.50
Weight (kg)	68.45	0.49	68.35	2.62	0.05
Sitting Height	91.30	0.99	86.05	6.58	1.12
Height Weight Ratio	43.18	0.64	42.29	1.01	1.05
Endomorphy	2.37	1.22	2.00	0.06	0.43
Mesomorphy	4.66	0.01	5.62	0.31	4.37*
Ectomorphy	3.03	0.47	2.37	0.73	1.07
Fat (%)	13.28	4.55	12.34	0.02	0.29

Table 11
Weight category wise morphological profile of Punjab and Himachal Pradesh STC boxers(Weight category 80 kg, N= 4)

Parameter	Mastuana (n=2)		Bilaspur (n=2)		Statistics
	Mean	SD	Mean	SD	t-value
Decimal Age (yrs.)	16.43	0.40	17.33	0.44	2.14*
Height (cm)	180.65	6.58	175.25	8.84	0.69
Weight (kg)	92.75	1.77	93.75	0.64	0.75
Sitting Height	89.00	8.20	81.10	2.40	1.31
Height Weight Ratio	40.48	0.23	40.50	2.44	0.01
Endomorphy	5.26	1.61	3.41	1.69	1.12
Mesomorphy	6.04	0.66	7.06	2.03	0.68
Ectomorphy	1.05	0.17	1.33	1.42	0.28
Fat (%)	23.55	4.76	17.73	5.52	1.13

Table 12
Morphological characteristics of Indian male boxers by GulshanKhanna and Indrani Manna (2006)

Parameter	Junior boxer (N=30)		Senior boxer (N=30)	
	Mean	SD	Mean	SD
Age (yrs)	17.6	2.9	22.1	3.1
Height (cm)	174	0.6	179	0.08
Weight (kg)	53.6	4.1	76.7	10.9
Endomorphy	1.8	0.5	2.3	0.6
Mesomorphy	3.2	0.6	4.9	0.7
Ectomorphy	4.0	0.8	2.3	0.8
Fat %	12.2	1.1	16.4	3.8

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